

COMP101

Introduction to Programming

2025-26

Lab-09

List Join and String Split

Objectives: No exercises – study slides only

List join

```
word_list = ["name one", "name two"]
```

```
print (word_list)
```

```
['name one', 'name two']
```

#o/p is a list – brackets and apostrophes

#o/p as a string with ten spaces between items: separator_string is ten spaces

```
word_list_string = ' '.join(word_list)
```

```
print (word_list_string)
```

```
name one      name two
```

#o/p is as a string with 10 spaces between the words

name one and name two are separated by ten spaces: We specified ten spaces in the separator_string:
The separator string can have any character in the quotes – the character chosen will o/p between the words

List join

```
word_list = ["name one", "name two"]
```

```
print (word_list)
```

```
['name one', 'name two']
```

#separator string is a comma

```
word_list_string = ",".join(word_list)
```

```
print (word_list_string)
```

```
name one, name two
```

`name one` and `name two` are separated by a comma because we specified a comma in the `separator_string`:

List join

#four items in the list

```
word_list = ["This", "is", "a", "sentence"]
```

```
word_list_string = ' '.join(word_list)
```

```
print (word_list_string)
```

This is a sentence

The separator_string is a space ' ' – put a space between the quotes

If we don't physically put a space in it: `word_list_string = ".join(word_list)`
the words (elements) in the list all join up: `Thisisasentence`

List join

```
word_list = ["This", "is", "a", "sentence"]
```

#separate the words with three asterix

```
word_list_string = "*** ".join(word_list)
```

```
print (word_list_string)
```

```
This*** is*** a*** sentence
```

Separator is three astertix

List join

```
word_list = ["This", "is", "a", "sentence"]
```

```
#separate the words with a tab (4 spaces)
```

```
word_list_string = "\t".join(word_list)
```

```
print (word_list_string)
```

```
This      is      a      sentence
```

Separator is a tab (4 spaces)

String split

```
name_string = "test name"  
name_list = name_string.split()  
print (name_list)  
['test', 'name']
```

The string is split on the space separating the string words
If not specified, it defaults to whitespace:
We could code it as:
`name_list = name_string.split(" ")`

```
nums_string = "name,name1,name2,name3"  
nums_list = nums_string.split(",")  
print (nums_list)  
['name', 'name1', 'name2', 'name3']
```

The split is on the comma separating the words in the string

The commas here are the list-separator commas

String split

```
sentence_string = "This is a sentence"  
sentence_list = sentence_string.split()
```

```
print (sentence_list)  
['This', 'is', 'a', 'sentence']
```

This is a single string: The words in it are not discrete units, i.e. we can't process individual words, only process the whole string

.split() takes the whole string and splits it on the spaces between words to produce a list of words, where each word in the original string can now be processed as an individual string in its own right

String split

Output all words in a string as a list

```
sentence_string = "This is a sentence"  
sentence_list = sentence_string.split()  
print (sentence_list)  
['This', 'is', 'a', 'sentence']
```

```
print (len(sentence_string))  
18  
print (len(sentence_list))  
4
```

We can't do things like sorting or reversing on this string, only basic processing, e.g. len() etc

We can show we have split the string:
len(sentence_string) counts the characters in the single string;

len(sentence_list) counts the words, i.e. strings, in the list that .split() has created

String split

Output a split on a specified substring

```
sentence_string = "This, is a, string"
```

```
sentence_list = sentence_string.split(",")  
print (sentence_list)
```

```
['This', ' is a', ' string']
```

These are not the commas from the original string – these are the list-separator commas

If the parentheses are empty, the split defaults to space between words:
Here we have added a parameter to split the string on every comma

String split

Output a split on a specified substring

```
sentence_string = "This, is a, string"
```

```
sentence_list = sentence_string.split("i")  
print (sentence_list)
```

```
['Th', 's, ', 's a, str', 'ng']
```

If the parentheses are empty, the split defaults to space between words:

Here we have added a parameter to split the string on every letter 'i'

A string can be split on any sequence character:

The default is to split on space between words or we can supply the split parameter.

A string can be split on `\n` or `\t` if in it i.e, split on the newline character or the tab character

String list () method

- Output a string as a list of its individual characters

```
sentence = "This is a sentence"  
sentence_list = list(sentence)
```

```
print (sentence_list)
```

```
['T', 'h', 'i', 's', ' ', 'i', 's', ' ', 'a', ' ', 's', 'e', 'n', 't', 'e', 'n', 'c', 'e']
```